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CSE - AIML

Company: WorkIndia

Job Profile: SDE-Intern

Offer Type: Internship + Placement

Internship Stipend: 40000

CTC: 15 Fixed + 1 Joining

Location: Bangalore

Process Date: 9th July, 2025

Placement Process:

Initially there was a Resume Shortlisting and a list was shared of candidates applicable to attend the interview process. The entire process was scheduled on 9th July itself from morning to evening and all interviews were conducted in an online mode.

1. Rapid Fire Round [72 students were present for this round]

As the name suggests, we had to attempt 40-50 questions in a 15 minute window so both accuracy and speed were essential. The questions were like : time/space complexity of code snippets, calculating the expected output of a given code and fundamental questions like if $a = 10$ then what will $a++ + ++a$ equal to - answer is 22. The snippets were a mix of Python,CPP and Java codes so cannot isolate one language to focus on. There were also CS fundamental questions like OS,CCN and Git commands.

2. Debugging Round [46 students appeared for this round]

We were sent a GMeet link where the interviewer asked for a choice between JS or Python. As I did DSA in Python and was aware of both Flask and FastAPI, I chose Python. I was a bit scared as the codebase shared was in Django so i told the interviewer honestly that I was aware of Flask and FastAPI so he asked if I wanted to switch to Express and JS or would I like to try Django as Google/StackOverflow/Documentation (everything except AI) was allowed so I stuck with Django and was able to debug the errors. Specific questions were

asked related to the errors so it is important to understand the error and explain your intuition as to why the error may have occurred and then solve it. General questions related to Git or the MVC folder structure in a codebase were also asked.

3. **DSA Round** [15 students appeared for this round]

This was the first time I was told to introduce myself, I was not prepared for this so always be prepared for this question else it sets an impression that you are not even prepared for the basic most obvious questions. I was lucky enough that this did not affect me. Overall the panelists asked candidates to only write pseudocode on Google Doc but i was expected to code and run it on GDB probably bcuz i code in Python so they wanted to see the output. There were 2 interviewers in the GMeet and I was asked this question :

In a gaming tournament unique ids are to be assigned from 1 to N to N participants, there might be some invalid ids already assigned in between and we need to assign them values acc to the next smallest valid number from 1 to N. So if the given array is 1,0,2 then 0 is invalid then answer should be 1,3,2. Another example test case :

Input : 2,0,4,-6,9,6

Output : 2,1,4,3,5,6

So my initial approach was to iterate over the array and put numbers already valid in a Set in Python which translates to a unordered_set in CPP. Now start a loop from $i = 1$ to N and check if it exists in the set , if not then replace the first invalid number in the array with the current i . This is $O(N)$ time and space. I was asked to reduce the space after this. I asked if the order matters while assigning values, so the interviewer cleared that the only expectation is 1 to N , nothing else matters. So my next approach was to sort the array and check if 1 to N exists in it with Binary Search(lower_bound), if not replace the first invalid position with the current missing number. This is $N\log N$ time and constant space. I was asked to bring time back to $O(N)$. I took a lot of time here and asked for hints where the interviewer told me try to establish some relation between indexes and validity of a number which eventually led me to a Cycle Sort solution where there is a special case that if the input is strictly between a given range like 1 to N then sorting can be done in $O(N)$. Pls read about this on GPT/Google, the solution goes as : We will try to place the already valid numbers on their correct indexes which will leave only invalid numbers on indexes where there should have been something else, now in another iteration from left to right we change the invalid numbers to valid because based on indexing we know the next valid number. I was able to code my initial 2 solutions but messed up something in the last one but the interviewer said that was ok.

Useful Tips :

- 1) Think everything out loud, explain your decisions like why did i take a Set in my initial approach
- 2) Be thorough with Time and Space Complexity
- 3) Ask unique questions : As one of my project was based on Kubernetes, i asked whether WorkIndia uses Kubernetes on premise or something like Cloud supported AWS EKS, the answer was AWS EKS.

4. DBMS Round [7 students appeared for this round]

I was again asked to introduce myself in brief and then asked to design a schema for an E-Commerce platforms with entities as - Users, Products, Orders and Reviews. There is no need to focus on what properties the User will have just give user_id as PK and move forward with the design. The important thing to handle here is the type of relationship - 1 to 1, 1 to N, M to N and design your tables in accordance with this and specify the foreign keys accurately. The interviewers were very friendly and at times I did take some poor design considerations and they immediately questioned that choice with a use case where my schema would have failed so I modelled accordingly and tried to keep the Schema error free. After that i was asked 2-3 SQL queries on my own schema. I took a lot of time and patience to solve this as 4-5 joins were required in each query so the important thing here is to keep talking about what you are thinking and why are you making a particular join. I was not able to design only one thing where i was asked to model categories like consider an Iphone with choices like size(small,medium,large), colour(red,green,blue) and RAM(4,8,16) so there are like 3^3 combinations possible so i want to store all this in MySQL lets say, i tried but could not reach a solution, so i offered a trick where i take the primary key from products and store these categories and their values as arrays in MongoDB. They told the solution is correct but can be done entirely in MySQL also to which i honestly replied i am not aware of that so in case you get stuck try to atleast give some solution rather than staying blank and silent. **Hackathon main kaisa thoda jugaad lagate hai to quickly achieve something waise kuch kar sakte ho, now this was accepted in WI which was good for me but i have no clue as to how other companies will perceive this so BE CAUTIOUS.**

Useful Tips :

- 1) Get SQL fundamentals sorted, preferably build an entire project with SQL as the DB and design schema yourself not by GPT, you will learn a lot and this round will be natural to you
- 2) Ask on-spot questions relevant to how the interview progressed : I had one unique question prepared for this so i kept that for later and instead first i asked : Whether WorkIndia uses a mixture of databases to achieve the required functionality ? This was due to the question where i shifted to MongoDB. Answer - Mixture of DB is used.
- 3) Ask unique questions : As WorkIndia will face a challenge of verifying the authenticity of Job Postings, to counter this does WI use a rule-based model or some AI algorithm or a decentralized approach with smart contracts on a blockchain. Answer - Rule based algorithms.

4. Culture Fit Round [4 students appeared for this round]

This round was like a roller coaster, too many ups and downs. Surprisingly, i was not asked to introduce myself, instead i was asked to give feedback of how the process felt to me. I answered honestly by saying the process can be considered as wholistic judgement for an SDE role where DSA,DBMS and Dev everything is evaluated. To be frank i was not prepared at all for HR questions, so i have answered with the first thing that come to my mind. As my interview was conducted by CTO of the company, i was asked Tech questions in between also. My interview was a mix of English and Hindi so no language enforcement as such but switch to Hindi only after the interviewer does it first. I will list all the next questions in order :

- 1) Why cant AI replace you - (Something similar to What are your Strengths or Why should we hire you)
- 2) Explain clustering in ML - (Probably bcuz I am from CSE - AIML)
- 3) Ghar pe kaun kaun rehta hai ? Mom Dad kya karte hai ? - (Basic answer)
- 4) Do u know how LLMs work ? Explain GenAI - (I was not aware so i told that honestly but i mentioned that my 7th Sem electives are focused on these topics. This kind of created a misunderstanding where the CTO thought I only learn a topic when the college teaches it but i cleared that up by explaining i was already working on Dev, DSA, CS Fundamentals, ML, Blockchain and Kubernetes so going into NLP and LLMs would be like Jack of all trades and master of none. It is very important to clear ur side without being overly defensive and most importantly being polite and non egoistic.

5) What inspired you to do Engineering and kabhi aisa laga kya ki sab chhod du - (Answer honestly, i told that pehle DSA start kiya tha tab lagta tha nhi ho raha merese but consistent solving, persistence and hard work eventually pays off)

6) How would you implement context switching in a GPU - (No idea, told about Context Switching in CPU and general idea about GPU like ML tasks ke liye better hai)

7) Any questions from your side - (Could you tell an idea about how the interns will be assigned teams and what projects we may work on, this was my ques)

2 students were selected after this round

Useful Tips:

1) Some preparation for HR questions can be done but be true to yourself and answer intuitively rather than creating fake scenarios

2) I have no clue as to why my Culture Fit Interview was so AIML focussed, might be because of my branch so no extra preparation is required for this but be prepared for surprise questions

Feel free to contact me on LinkedIn - <https://www.linkedin.com/in/ninad-maadhavi/>

Resume I used for the placement season -  Ninad_Maadhavi_CSE.pdf